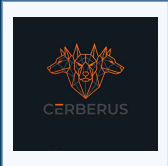


ESG Factor Backtest



INTERNAL

Backtest ESG score factor returns with sector-neutral quintile construction.

Author: Cerberus Research

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N	12,692
CLASSIFICATION	internal
AUTHOR	Cerberus Research
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DURATION	39.5s
DATASETS	fundamentals/fundamentals_esg, fundamentals/fundamentals_curated
PARAMETER · LOOKBACK_YEARS	5
PARAMETER · HORIZON	63
PARAMETER · TOP_N	50
KPI	Value
N	12,692

KEY FINDINGS

- Symbols with ESG signal: 782
- ESG portfolio annualized return: 0.1% (SPY: 0.1%)

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- 02 Results & Interpretation
- 03 Key Findings
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Generated by the Cerberus studies framework · schema 2

Data Sources

- fundamentals/fundamentals_esg
- fundamentals/fundamentals_curated

Key Findings

- Symbols with ESG signal: 782
- ESG portfolio annualized return: 0.1% (SPY: 0.1%)

Executive Summary

NOTE

Backtests an ESG factor by constructing an equal-weighted ESG-flagged portfolio and benchmarking it against SPY. When granular ESG scores are available, sector-neutral quintile forward returns are also computed.

Results & Interpretation

NOTE

Backtest ESG score factor returns with sector-neutral quintile construction.

The study analysed 12,692 rows of data.

Main takeaways:

- Symbols with ESG signal: 782
- ESG portfolio annualized return: 0.1% (SPY: 0.1%)

Objectives

NOTE

Evaluate whether the studied signal produces a measurable, repeatable edge.

Quantify the reliability of the observed patterns across the sample.

Methodology

NOTE

Analysis is run on the configured datasets with the parameters listed in the Study Configuration block. Results are aggregated into summary statistics, tables, and visual one-pagers for review.

Charts



Figure 1. ESG portfolio vs SPY cumulative returns
ESG portfolio vs SPY cumulative returns



Figure 2. ESG factor backtest dashboard
ESG factor backtest dashboard

Statistical Results

symbol	snapshot_mo...	sector	esgScore
1SIE.MI	2026-07	Industrials	1
A	2026-06	Healthcare	1
AAL	2026-06	Industrials	1
AAPL	2026-06	Technology	1
ABBV	2026-06	Healthcare	1

symbol	snapshot_mo...	sector	esgScore
ABN.AS	2026-07	Financial Services	1
ABNB	2026-06	Consumer Cyclical	1
ABT	2026-06	Healthcare	1
ACA.PA	2026-07	Financial Services	1
ACGL	2026-06	Financial Services	1
ACHR	2026-06	Industrials	1
ACLS	2026-06	Technology	1
ACM	2026-06	Industrials	1
ACN	2026-06	Technology	1
ACS.MC	2026-07	Industrials	1
ACX.MC	2026-07	Basic Materials	1
AD.AS	2026-07	Consumer Defensive	1
ADBE	2026-06	Technology	1
ADI	2026-06	Technology	1
ADM	2026-06	Consumer Defensive	1
ADP	2026-06	Technology	1
ADS.DE	2026-07	Consumer Cyclical	1
ADSK	2026-06	Technology	1
AEE	2026-06	Utilities	1
AEM	2026-06	Basic Materials	1
AENA.MC	2026-07	Industrials	1
AEP	2026-06	Utilities	1
AES	2026-06	Utilities	1
AFL	2026-06	Financial Services	1
AFRM	2026-06	Financial Services	1
AGI	2026-06	Basic Materials	1
AIG	2026-06	Financial Services	1
AIR.DE	2026-07	Industrials	1
AIR.PA	2026-07	Industrials	1
AIZ	2026-06	Financial Services	1
AJG	2026-06	Financial Services	1
AKAM	2026-06	Technology	1

Research only. Not investment advice. Past performance is not indicative of future results.

symbol	snapshot_mo...	sector	esgScore
AKZA.AS	2026-07	Basic Materials	1
ALB	2026-06	Basic Materials	1
ALGN	2026-06	Healthcare	1
ALK	2026-06	Industrials	1
ALL	2026-06	Financial Services	1
ALLE	2026-06	Industrials	1
ALNY	2026-06	Healthcare	1
ALT	2026-06	Healthcare	1
ALV.DE	2026-07	Financial Services	1
AMAT	2026-06	Technology	1
AMCR	2026-06	Consumer Cyclical	1
AMD	2026-06	Technology	1
AME	2026-06	Industrials	1
AMGN	2026-06	Healthcare	1
AMKR	2026-06	Technology	1
AMP	2026-06	Financial Services	1
AMP.MI	2026-07	Healthcare	1
AMS.MC	2026-07	Technology	1
AMSC	2026-06	Industrials	1
AMT	2026-06	Real Estate	1
AMZN	2026-06	Consumer Cyclical	1
ANDE	2026-06	Consumer Defensive	1
ANET	2026-06	Technology	1
AON	2026-06	Financial Services	1
AOS	2026-06	Industrials	1
APA	2026-06	Energy	1
APD	2026-06	Basic Materials	1
APH	2026-06	Technology	1
APLD	2026-06	Technology	1
APO	2026-06	Financial Services	1
APP	2026-06	Communication Services	1
APPF	2026-06	Technology	1

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symbol	snapshot_mo...	sector	esgScore
APPN	2026-06	Technology	1
APPS	2026-06	Technology	1
APTV	2026-06	Consumer Cyclical	1
ARCO	2026-06	Consumer Cyclical	1
ARE	2026-06	Real Estate	1
ARES	2026-06	Financial Services	1
ARGX	2026-06	Healthcare	1
ARM	2026-06	Technology	1
ARWR	2026-06	Healthcare	1
ASM.AS	2026-07	Technology	1
ASML	2026-06	Technology	1
ASML.AS	2026-07	Technology	1
ASPN	2026-06	Industrials	1
ASTS	2026-06	Technology	1
ATAI	2026-06	Healthcare	1
ATAT	2026-06	Consumer Cyclical	1
ATEN	2026-06	Technology	1
ATO	2026-06	Utilities	1
AVB	2026-06	Real Estate	1
AVGO	2026-06	Technology	1
AVO	2026-06	Consumer Defensive	1
AVY	2026-06	Consumer Cyclical	1
AWK	2026-06	Utilities	1
AWR	2026-06	Utilities	1
AXON	2026-06	Industrials	1
AXP	2026-06	Financial Services	1
AZM.MI	2026-07	Financial Services	1
AZN	2026-06	Healthcare	1
AZO	2026-06	Consumer Cyclical	1
B	2026-06	Basic Materials	1
BA	2026-06	Industrials	1
BABA	2026-06	Consumer Cyclical	1

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symbol	snapshot_mo...	sector	esgScore
BAC	2026-06	Financial Services	1
BAH	2026-06	Industrials	1
BALL	2026-06	Consumer Cyclical	1
BAMI.MI	2026-07	Financial Services	1
BAS.DE	2026-07	Basic Materials	1
BAX	2026-06	Healthcare	1
BAYN.DE	2026-07	Healthcare	1
BBAI	2026-06	Technology	1
BBVA.MC	2026-07	Financial Services	1
BBWI	2026-06	Consumer Cyclical	1
BBY	2026-06	Consumer Cyclical	1
BCP.LS	2026-07	Financial Services	1
BDX	2026-06	Healthcare	1
BEAM	2026-06	Healthcare	1
BEN	2026-06	Financial Services	1
BF-B	2026-06	Consumer Defensive	1
BG	2026-06	Consumer Defensive	1
BGI	2026-06	Consumer Cyclical	1
BIDU	2026-06	Communication Services	1
BIIB	2026-06	Healthcare	1
BIO	2026-06	Healthcare	1
BK	2026-06	Financial Services	1
BKNG	2026-06	Consumer Cyclical	1
BKR	2026-06	Energy	1
BKSY	2026-07	Industrials	1
BKT.MC	2026-07	Financial Services	1
BLDR	2026-06	Industrials	1
BLK	2026-06	Financial Services	1
BMED.MI	2026-07	Financial Services	1
BMRN	2026-06	Healthcare	1
BMW.DE	2026-07	Consumer Cyclical	1
BMY	2026-06	Healthcare	1

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symbol	snapshot_mo...	sector	esgScore
BNP.PA	2026-07	Financial Services	1
BPE.MI	2026-07	Financial Services	1
BR	2026-06	Technology	1
BRK-B	2026-06	Financial Services	1
BRO	2026-06	Financial Services	1
BROS	2026-06	Consumer Cyclical	1
BSP	2026-07	Technology	1
BSX	2026-06	Healthcare	1
BULL	2026-06	Technology	1
BWA	2026-06	Consumer Cyclical	1
BWXT	2026-06	Industrials	1
BX	2026-06	Financial Services	1
BXP	2026-06	Real Estate	1
C	2026-06	Financial Services	1
CABK.MC	2026-07	Financial Services	1
CACI	2026-06	Technology	1
CAG	2026-06	Consumer Defensive	1
CAH	2026-06	Healthcare	1
CARR	2026-06	Industrials	1
CASY	2026-06	Consumer Cyclical	1
CAT	2026-06	Industrials	1
CB	2026-06	Financial Services	1
CBOE	2026-06	Financial Services	1
CBRE	2026-06	Real Estate	1
CCI	2026-06	Real Estate	1
CCJ	2026-06	Energy	1
CCL	2026-06	Consumer Cyclical	1
CDNS	2026-06	Technology	1
CDW	2026-06	Technology	1
CE	2026-06	Basic Materials	1
CEG	2026-07	Utilities	1
CELH	2026-07	Consumer Defensive	1

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symbol	snapshot_mo...	sector	esgScore
CF	2026-07	Basic Materials	1
CFG	2026-07	Financial Services	1
CGC	2026-07	Healthcare	1
CGNT	2026-07	Technology	1
CHD	2026-07	Consumer Defensive	1
CHDN	2026-07	Consumer Cyclical	1
CHKP	2026-07	Technology	1
CHRW	2026-07	Industrials	1
CHTR	2026-07	Communication Services	1
CI	2026-07	Healthcare	1
CIEN	2026-07	Technology	1
CIFR	2026-07	Technology	1
CINF	2026-07	Financial Services	1
CL	2026-07	Consumer Defensive	1
CLH	2026-07	Industrials	1
CLNX.MC	2026-07	Real Estate	1
CLX	2026-07	Consumer Defensive	1
CMBT	2026-07	Energy	1
CMCSA	2026-07	Communication Services	1
CME	2026-07	Financial Services	1
CMG	2026-07	Consumer Cyclical	1
CMI	2026-07	Industrials	1
CMPS	2026-07	Healthcare	1
CMRE	2026-07	Industrials	1
CMS	2026-07	Utilities	1
CNC	2026-07	Healthcare	1
CNI	2026-07	Industrials	1
CNP	2026-07	Utilities	1
COCO	2026-07	Consumer Defensive	1
COF	2026-07	Financial Services	1
COHR	2026-07	Technology	1
COIN	2026-07	Financial Services	1

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symbol	snapshot_mo...	sector	esgScore
COL.MC	2026-07	Real Estate	1
CON.DE	2026-07	Consumer Cyclical	1
COO	2026-07	Healthcare	1
COP	2026-07	Energy	1
CORZ	2026-07	Technology	1
COST	2026-07	Consumer Defensive	1
CPAY	2026-07	Technology	1
CPB	2026-07	Consumer Defensive	1
CPR.MI	2026-07	Consumer Defensive	1
CPRT	2026-07	Industrials	1
CPT	2026-07	Real Estate	1
CRCL	2026-07	Financial Services	1
CRH	2026-07	Basic Materials	1
CRL	2026-07	Healthcare	1
CRM	2026-07	Technology	1
CRON	2026-07	Healthcare	1
CRSP	2026-07	Healthcare	1
CRWD	2026-07	Technology	1
CRWV	2026-07	Technology	1
CSCO	2026-07	Technology	1
CSGP	2026-07	Real Estate	1
CSV	2026-07	Consumer Cyclical	1
CSX	2026-07	Industrials	1
CTAS	2026-07	Industrials	1
CTSH	2026-07	Technology	1
CTVA	2026-07	Basic Materials	1
CVLT	2026-07	Technology	1
CVNA	2026-07	Consumer Cyclical	1
CVS	2026-07	Healthcare	1
CVX	2026-07	Energy	1
CWCO	2026-07	Utilities	1
CWST	2026-07	Industrials	1

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symbol	snapshot_mo...	sector	esgScore
CWT	2026-07	Utilities	1
CZR	2026-07	Consumer Cyclical	1
D	2026-07	Utilities	1
DAC	2026-07	Industrials	1
DAL	2026-07	Industrials	1
DASH	2026-07	Consumer Cyclical	1
DB1.DE	2026-07	Financial Services	1
DBK.DE	2026-07	Financial Services	1
DD	2026-07	Basic Materials	1
DDOG	2026-07	Technology	1
DE	2026-07	Industrials	1
DECK	2026-07	Consumer Cyclical	1
DELL	2026-07	Technology	1
DG	2026-07	Consumer Defensive	1
DGX	2026-07	Healthcare	1
DHI	2026-07	Consumer Cyclical	1
DHL.DE	2026-07	Industrials	1
DHR	2026-07	Healthcare	1
DHT	2026-07	Energy	1
DIA.MI	2026-07	Healthcare	1
DINO	2026-07	Energy	1
DIS	2026-07	Communication Services	1
DJT	2026-07	Communication Services	1
DKNG	2026-07	Consumer Cyclical	1
DLR	2026-07	Real Estate	1
DLTR	2026-07	Consumer Defensive	1
DNN	2026-07	Energy	1
DOC	2026-07	Real Estate	1
DOV	2026-07	Industrials	1
DOW	2026-07	Basic Materials	1
DPZ	2026-07	Consumer Cyclical	1
DRI	2026-07	Consumer Cyclical	1

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symbol	snapshot_mo...	sector	esgScore
DSY.PA	2026-07	Technology	1
DTE	2026-07	Utilities	1
DTE.DE	2026-07	Communication Services	1
DTG.DE	2026-07	Industrials	1
DUK	2026-07	Utilities	1
DUOL	2026-07	Technology	1
DVA	2026-07	Healthcare	1
DVN	2026-07	Energy	1
DXCM	2026-07	Healthcare	1
EA	2026-07	Communication Services	1
EBAY	2026-07	Consumer Cyclical	1
ECL	2026-07	Basic Materials	1
ED	2026-07	Utilities	1
EFX	2026-07	Industrials	1
EG	2026-07	Financial Services	1
EIX	2026-07	Utilities	1
EL	2026-07	Consumer Defensive	1
ELE.MC	2026-07	Utilities	1
ELF	2026-07	Consumer Defensive	1
ELMD	2026-07	Healthcare	1
ELV	2026-07	Healthcare	1
EME	2026-07	Industrials	1
EMN	2026-07	Basic Materials	1
EMR	2026-07	Industrials	1
EN.PA	2026-07	Industrials	1
ENG.MC	2026-07	Utilities	1
ENI.MI	2026-07	Energy	1
ENPH	2026-07	Technology	1
ENR.DE	2026-07	Industrials	1
ENTG	2026-07	Technology	1
EOAN.DE	2026-07	Utilities	1
EOG	2026-07	Energy	1

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symbol	snapshot_mo...	sector	esgScore
EPAM	2026-07	Technology	1
EQIX	2026-07	Real Estate	1
EQR	2026-07	Real Estate	1
EQT	2026-07	Energy	1
ERIE	2026-07	Financial Services	1
ES	2026-07	Utilities	1
ESS	2026-07	Real Estate	1
ESTC	2026-07	Technology	1
ETN	2026-07	Industrials	1
ETR	2026-07	Utilities	1
ETSY	2026-07	Consumer Cyclical	1
EVRG	2026-07	Utilities	1
EW	2026-07	Healthcare	1
EXC	2026-07	Utilities	1
EXE	2026-07	Energy	1
EXEL	2026-07	Healthcare	1
EXPD	2026-07	Industrials	1
EXPE	2026-07	Consumer Cyclical	1
EXR	2026-07	Real Estate	1
F	2026-07	Consumer Cyclical	1
FANG	2026-07	Energy	1
FAST	2026-07	Industrials	1
FCX	2026-07	Basic Materials	1
FDS	2026-07	Financial Services	1
FDX	2026-07	Industrials	1
FE	2026-07	Utilities	1
FER	2026-07	Industrials	1
FFIV	2026-07	Technology	1
FICO	2026-07	Technology	1
FIS	2026-07	Technology	1
FISV	2026-07	Unknown	1
FITB	2026-07	Financial Services	1

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symbol	snapshot_mo...	sector	esgScore
FIX	2026-07	Industrials	1
FMC	2026-07	Basic Materials	1
FNV	2026-07	Basic Materials	1
FOX	2026-07	Communication Services	1
FOXA	2026-07	Communication Services	1
FRO	2026-07	Energy	1
FRT	2026-07	Real Estate	1
FSLR	2026-07	Technology	1
FTNT	2026-07	Technology	1
FTV	2026-07	Technology	1
GCT	2026-07	Technology	1
GD	2026-07	Industrials	1
GDDY	2026-07	Technology	1
GE	2026-07	Industrials	1
GEHC	2026-07	Healthcare	1
GEN	2026-07	Technology	1
GEO	2026-07	Industrials	1
GEV	2026-07	Industrials	1
GFS	2026-07	Technology	1
GILD	2026-07	Healthcare	1
GIS	2026-07	Consumer Defensive	1
GL	2026-07	Financial Services	1
GLW	2026-07	Technology	1
GM	2026-07	Consumer Cyclical	1
GNK	2026-07	Industrials	1
GNRC	2026-07	Industrials	1
GOLD	2026-07	Financial Services	1
GOOG	2026-07	Communication Services	1
GOOGL	2026-07	Communication Services	1
GPC	2026-07	Consumer Cyclical	1
GPN	2026-07	Industrials	1
GRMN	2026-07	Technology	1

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symbol	snapshot_mo...	sector	esgScore
GS	2026-07	Financial Services	1
GWW	2026-07	Industrials	1
HAL	2026-07	Energy	1
HAS	2026-07	Consumer Cyclical	1
HBAN	2026-07	Financial Services	1
HCA	2026-07	Healthcare	1
HD	2026-07	Consumer Cyclical	1
HIG	2026-07	Financial Services	1
HII	2026-07	Industrials	1
HIMS	2026-07	Healthcare	1
HLT	2026-07	Consumer Cyclical	1
HON	2026-07	Industrials	1
HOOD	2026-07	Financial Services	1
HPE	2026-07	Technology	1
HPQ	2026-07	Unknown	1
HRL	2026-07	Consumer Defensive	1
HSIC	2026-07	Healthcare	1
HST	2026-07	Real Estate	1
HSY	2026-07	Consumer Defensive	1
HUBB	2026-07	Industrials	1
HUBS	2026-07	Technology	1
HUM	2026-07	Healthcare	1
HUT	2026-07	Financial Services	1
HWM	2026-07	Industrials	1
IBKR	2026-07	Financial Services	1
IBM	2026-07	Technology	1
ICE	2026-07	Financial Services	1
IDXX	2026-07	Healthcare	1
IESC	2026-07	Industrials	1
IEX	2026-07	Industrials	1
IFF	2026-07	Basic Materials	1
ILMN	2026-07	Healthcare	1

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symbol	snapshot_mo...	sector	esgScore
INCY	2026-07	Healthcare	1
INSM	2026-07	Healthcare	1
INSW	2026-07	Energy	1
INTC	2026-07	Technology	1
INTU	2026-07	Technology	1
INVH	2026-07	Real Estate	1
IONQ	2026-07	Technology	1
IP	2026-07	Consumer Cyclical	1
IQV	2026-07	Healthcare	1
IR	2026-07	Industrials	1
IREN	2026-07	Financial Services	1
IRM	2026-07	Real Estate	1
IRON	2026-07	Healthcare	1
IRWD	2026-07	Healthcare	1
ISRG	2026-07	Healthcare	1
IT	2026-07	Technology	1
ITW	2026-07	Industrials	1
IVZ	2026-07	Financial Services	1
J	2026-07	Industrials	1
JAZZ	2026-07	Healthcare	1
JBHT	2026-07	Industrials	1
JBL	2026-07	Technology	1
JCI	2026-07	Industrials	1
JD	2026-07	Consumer Cyclical	1
JKHY	2026-07	Technology	1
JNJ	2026-07	Healthcare	1
JOBY	2026-07	Industrials	1
JPM	2026-07	Financial Services	1
KDP	2026-07	Consumer Defensive	1
KEX	2026-07	Industrials	1
KEY	2026-07	Financial Services	1
KEYS	2026-07	Technology	1

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symbol	snapshot_mo...	sector	esgScore
KGC	2026-07	Basic Materials	1
KHC	2026-07	Consumer Defensive	1
KIM	2026-07	Real Estate	1
KKR	2026-07	Financial Services	1
KLAC	2026-07	Technology	1
KMB	2026-07	Consumer Defensive	1
KMI	2026-07	Energy	1
KMX	2026-07	Consumer Cyclical	1
KO	2026-07	Consumer Defensive	1
KR	2026-07	Unknown	1
KTOS	2026-07	Industrials	1
KULR	2026-07	Technology	1
KVUE	2026-07	Consumer Defensive	1
L	2026-07	Financial Services	1
LCID	2026-07	Consumer Cyclical	1
LCTX	2026-07	Healthcare	1
LDOS	2026-07	Technology	1
LEN	2026-07	Consumer Cyclical	1
LEU	2026-07	Energy	1
LH	2026-07	Healthcare	1
LHX	2026-07	Industrials	1
LI	2026-07	Consumer Cyclical	1
LII	2026-07	Industrials	1
LIN	2026-07	Basic Materials	1
LITE	2026-07	Technology	1
LKQ	2026-07	Consumer Cyclical	1
LLY	2026-07	Healthcare	1
LMT	2026-07	Industrials	1
LNG	2026-07	Energy	1
LNT	2026-07	Utilities	1
LOW	2026-07	Consumer Cyclical	1
LRCX	2026-07	Technology	1

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symbol	snapshot_mo...	sector	esgScore
LULU	2026-07	Consumer Cyclical	1
LUNR	2026-07	Industrials	1
LUV	2026-07	Industrials	1
LVS	2026-07	Consumer Cyclical	1
LW	2026-07	Consumer Defensive	1
LYB	2026-07	Basic Materials	1
LYV	2026-07	Communication Services	1
MA	2026-07	Financial Services	1
MAA	2026-07	Real Estate	1
MAR	2026-07	Consumer Cyclical	1
MARA	2026-07	Financial Services	1
MAS	2026-07	Industrials	1
MATX	2026-07	Industrials	1
MCD	2026-07	Consumer Cyclical	1
MCHP	2026-07	Technology	1
MCK	2026-07	Healthcare	1
MCO	2026-07	Financial Services	1
MDB	2026-07	Technology	1
MDLZ	2026-07	Consumer Defensive	1
MDT	2026-07	Healthcare	1
MELI	2026-07	Consumer Cyclical	1
MET	2026-07	Financial Services	1
META	2026-07	Communication Services	1
MGM	2026-07	Consumer Cyclical	1
MHK	2026-07	Consumer Cyclical	1
MKC	2026-07	Consumer Defensive	1
MKSI	2026-07	Technology	1
MKTX	2026-07	Financial Services	1
MLM	2026-07	Basic Materials	1
MMM	2026-07	Industrials	1
MNDY	2026-07	Technology	1
MNKD	2026-07	Healthcare	1

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symbol	snapshot_mo...	sector	esgScore
MNST	2026-07	Consumer Defensive	1
MO	2026-07	Consumer Defensive	1
MOD	2026-07	Consumer Cyclical	1
MOG-A	2026-07	Industrials	1
MOH	2026-07	Healthcare	1
MOS	2026-07	Basic Materials	1
MPC	2026-07	Energy	1
MPWR	2026-07	Technology	1
MRK	2026-07	Healthcare	1
MRNA	2026-07	Healthcare	1
MRSB	2026-07	Financial Services	1
MRVL	2026-07	Technology	1
MS	2026-07	Financial Services	1
MSCI	2026-07	Financial Services	1
MSEX	2026-07	Utilities	1
MSFT	2026-07	Technology	1
MSI	2026-07	Technology	1
MSTR	2026-07	Technology	1
MTB	2026-07	Financial Services	1
MTCH	2026-07	Communication Services	1
MTD	2026-07	Healthcare	1
MTZ	2026-07	Industrials	1
MU	2026-07	Technology	1
NABL	2026-07	Technology	1
NAT	2026-07	Energy	1
NBIS	2026-07	Communication Services	1
NBIX	2026-07	Healthcare	1
NCLH	2026-07	Consumer Cyclical	1
NDAQ	2026-07	Financial Services	1
NDSN	2026-07	Industrials	1
NEE	2026-07	Utilities	1
NEM	2026-07	Basic Materials	1

symbol	snapshot_mo...	sector	esgScore
NET	2026-07	Technology	1
NFLX	2026-07	Communication Services	1
NI	2026-07	Utilities	1
NIO	2026-07	Consumer Cyclical	1
NKE	2026-07	Consumer Cyclical	1
NMM	2026-07	Industrials	1
NOC	2026-07	Industrials	1
NOK	2026-07	Technology	1
NOW	2026-07	Technology	1
NRG	2026-07	Utilities	1
NSC	2026-07	Industrials	1
NSSC	2026-07	Industrials	1
NTAP	2026-07	Technology	1
NTCT	2026-07	Technology	1
NTR	2026-07	Basic Materials	1
NTRS	2026-07	Financial Services	1
NU	2026-07	Financial Services	1
NUE	2026-07	Basic Materials	1
NVDA	2026-07	Technology	1
NVO	2026-07	Healthcare	1
NVR	2026-07	Consumer Cyclical	1
NVT	2026-07	Industrials	1
NWS	2026-07	Communication Services	1
NWSA	2026-07	Communication Services	1
NXE	2026-07	Energy	1
NXPI	2026-07	Technology	1
O	2026-07	Real Estate	1
ODFL	2026-07	Industrials	1
OGI	2026-07	Healthcare	1
OKE	2026-07	Energy	1
OKLO	2026-07	Utilities	1
OKTA	2026-07	Technology	1

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symbol	snapshot_mo...	sector	esgScore
OLN	2026-07	Basic Materials	1
OMAB	2026-07	Industrials	1
OMC	2026-07	Communication Services	1
ON	2026-07	Technology	1
ONDS	2026-07	Technology	1
OPEN	2026-07	Real Estate	1
ORCL	2026-07	Technology	1
ORLY	2026-07	Consumer Cyclical	1
OSPN	2026-07	Technology	1
OTIS	2026-07	Industrials	1
OXY	2026-07	Energy	1
PAAS	2026-07	Basic Materials	1
PANW	2026-07	Technology	1
PARR	2026-07	Energy	1
PATH	2026-07	Technology	1
PAYC	2026-07	Technology	1
PAYX	2026-07	Technology	1
PCAR	2026-07	Industrials	1
PCG	2026-07	Utilities	1
PDD	2026-07	Consumer Cyclical	1
PEG	2026-07	Utilities	1
PEP	2026-07	Consumer Defensive	1
PFE	2026-07	Healthcare	1
PFG	2026-07	Financial Services	1
PG	2026-07	Consumer Defensive	1
PGR	2026-07	Financial Services	1
PH	2026-07	Industrials	1
PHM	2026-07	Consumer Cyclical	1
PINS	2026-07	Communication Services	1
PKG	2026-07	Consumer Cyclical	1
PL	2026-07	Industrials	1
PLD	2026-07	Real Estate	1

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symbol	snapshot_mo...	sector	esgScore
PLTR	2026-07	Technology	1
PM	2026-07	Consumer Defensive	1
PNC	2026-07	Financial Services	1
PNR	2026-07	Industrials	1
PNW	2026-07	Utilities	1
PODD	2026-07	Healthcare	1
PONY	2026-07	Technology	1
POOL	2026-07	Industrials	1
PPG	2026-07	Basic Materials	1
PPL	2026-07	Utilities	1
PRU	2026-07	Financial Services	1
PSA	2026-07	Real Estate	1
PSKY	2026-07	Communication Services	1
PSN	2026-07	Technology	1
PSX	2026-07	Energy	1
PTC	2026-07	Technology	1
PWR	2026-07	Industrials	1
Q	2026-07	Technology	1
QBTS	2026-07	Technology	1
QCOM	2026-07	Technology	1
QLYS	2026-07	Technology	1
QRVO	2026-07	Technology	1
RAMP	2026-07	Technology	1
RBLX	2026-07	Communication Services	1
RBRK	2026-07	Technology	1
RCAT	2026-07	Industrials	1
RCL	2026-07	Consumer Cyclical	1
RDWR	2026-07	Technology	1
REG	2026-07	Real Estate	1
REGN	2026-07	Healthcare	1
RF	2026-07	Financial Services	1
RGTI	2026-07	Technology	1

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symbol	snapshot_mo...	sector	esgScore
RHI	2026-07	Industrials	1
RIOT	2026-07	Financial Services	1
RJF	2026-07	Financial Services	1
RKLB	2026-07	Industrials	1
RKT	2026-07	Financial Services	1
RL	2026-07	Consumer Cyclical	1
RMD	2026-07	Healthcare	1
ROK	2026-07	Industrials	1
ROL	2026-07	Consumer Cyclical	1
ROP	2026-07	Technology	1
ROST	2026-07	Consumer Cyclical	1
RPD	2026-07	Technology	1
RSG	2026-07	Industrials	1
RTX	2026-07	Industrials	1
RVTY	2026-07	Healthcare	1
RXXRX	2026-07	Healthcare	1
RYTM	2026-07	Healthcare	1
S	2026-07	Technology	1
SAIC	2026-07	Technology	1
SAIL	2026-07	Technology	1
SATS	2026-07	Communication Services	1
SBAC	2026-07	Real Estate	1
SBLK	2026-07	Industrials	1
SBUX	2026-07	Consumer Cyclical	1
SCHW	2026-07	Financial Services	1
SCI	2026-07	Consumer Cyclical	1
SEB	2026-07	Industrials	1
SEDG	2026-07	Technology	1
SERV	2026-07	Industrials	1
SFL	2026-07	Industrials	1
SHOP	2026-07	Technology	1
SHW	2026-07	Basic Materials	1

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symbol	snapshot_mo...	sector	esgScore
SIRI	2026-07	Communication Services	1
SJM	2026-07	Consumer Defensive	1
SLB	2026-07	Energy	1
SMCI	2026-07	Technology	1
SMMT	2026-07	Healthcare	1
SMR	2026-07	Industrials	1
SNA	2026-07	Industrials	1
SNDK	2026-07	Technology	1
SNDL	2026-07	Consumer Defensive	1
SNOW	2026-07	Technology	1
SNPS	2026-07	Technology	1
SO	2026-07	Utilities	1
SOFI	2026-07	Financial Services	1
SOLV	2026-07	Healthcare	1
SPCX	2026-07	Industrials	1
SPG	2026-07	Real Estate	1
SPGI	2026-07	Financial Services	1
SRE	2026-07	Utilities	1
STE	2026-07	Healthcare	1
STLD	2026-07	Basic Materials	1
STNG	2026-07	Energy	1
STT	2026-07	Financial Services	1
STX	2026-07	Technology	1
STZ	2026-07	Consumer Defensive	1
SW	2026-07	Consumer Cyclical	1
SWBI	2026-07	Industrials	1
SWK	2026-07	Industrials	1
SWKS	2026-07	Technology	1
SYF	2026-07	Financial Services	1
SYK	2026-07	Healthcare	1
SYM	2026-07	Industrials	1
YYY	2026-07	Consumer Defensive	1

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symbol	snapshot_mo...	sector	esgScore
T	2026-07	Communication Services	1
TAP	2026-07	Consumer Defensive	1
TBPH	2026-07	Healthcare	1
TCOM	2026-07	Consumer Cyclical	1
TDG	2026-07	Industrials	1
TDY	2026-07	Technology	1
TEAM	2026-07	Technology	1
TECH	2026-07	Healthcare	1
TEL	2026-07	Technology	1
TEM	2026-07	Healthcare	1
TENB	2026-07	Technology	1
TER	2026-07	Technology	1
TFC	2026-07	Financial Services	1
TFX	2026-07	Healthcare	1
TGT	2026-07	Consumer Defensive	1
TJX	2026-07	Consumer Cyclical	1
TK	2026-07	Energy	1
TKO	2026-07	Communication Services	1
TLRY	2026-07	Healthcare	1
TLS	2026-07	Technology	1
TME	2026-07	Communication Services	1
TMO	2026-07	Healthcare	1
TMUS	2026-07	Communication Services	1
TNK	2026-07	Energy	1
TPL	2026-07	Energy	1
TPR	2026-07	Consumer Cyclical	1
TRGP	2026-07	Energy	1
TRI	2026-07	Industrials	1
TRMB	2026-07	Technology	1
TROW	2026-07	Financial Services	1
TRV	2026-07	Financial Services	1
TSCO	2026-07	Consumer Cyclical	1

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symbol	snapshot_mo...	sector	esgScore
TSLA	2026-07	Consumer Cyclical	1
TSM	2026-07	Technology	1
TSN	2026-07	Consumer Defensive	1
TT	2026-07	Industrials	1
TTD	2026-07	Communication Services	1
TTWO	2026-07	Communication Services	1
TXN	2026-07	Technology	1
TXT	2026-07	Industrials	1
TYL	2026-07	Technology	1
UAL	2026-07	Industrials	1
UBER	2026-07	Technology	1
UEC	2026-07	Energy	1
UHS	2026-07	Healthcare	1
ULTA	2026-07	Consumer Cyclical	1
UNH	2026-07	Healthcare	1
UNP	2026-07	Industrials	1
UPS	2026-07	Industrials	1
URI	2026-07	Industrials	1
USB	2026-07	Financial Services	1
UUUU	2026-07	Energy	1
V	2026-07	Financial Services	1
VFC	2026-07	Consumer Cyclical	1
VFF	2026-07	Consumer Defensive	1
VICI	2026-07	Real Estate	1
VIK	2026-07	Consumer Cyclical	1
VLO	2026-07	Energy	1
VLTO	2026-07	Industrials	1
VMC	2026-07	Basic Materials	1
VRNS	2026-07	Technology	1
VRSK	2026-07	Industrials	1
VRSN	2026-07	Technology	1
VRT	2026-07	Industrials	1

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symbol	snapshot_mo...	sector	esgScore
VRTX	2026-07	Healthcare	1
VST	2026-07	Utilities	1
VTR	2026-07	Real Estate	1
VTRS	2026-07	Healthcare	1
VZ	2026-07	Communication Services	1
WAB	2026-07	Industrials	1
WAT	2026-07	Healthcare	1
WBD	2026-07	Communication Services	1
WDAY	2026-07	Technology	1
WDC	2026-07	Technology	1
WEC	2026-07	Utilities	1
WELL	2026-07	Real Estate	1
WEN	2026-07	Consumer Cyclical	1
WFC	2026-07	Financial Services	1
WHR	2026-07	Consumer Cyclical	1
WM	2026-07	Industrials	1
WMB	2026-07	Energy	1
WMT	2026-07	Consumer Defensive	1
WPM	2026-07	Basic Materials	1
WRB	2026-07	Financial Services	1
WSM	2026-07	Consumer Cyclical	1
WST	2026-07	Healthcare	1
WTRG	2026-07	Utilities	1
WTW	2026-07	Financial Services	1
WULF	2026-07	Financial Services	1
WY	2026-07	Real Estate	1
WYNN	2026-07	Consumer Cyclical	1
XEL	2026-07	Utilities	1
XOM	2026-07	Energy	1
XPEV	2026-07	Consumer Cyclical	1
XRAY	2026-07	Healthcare	1
XYL	2026-07	Industrials	1

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symbol	snapshot_mo...	sector	esgScore
XYZ	2026-07	Technology	1
YORW	2026-07	Utilities	1
YUM	2026-07	Consumer Cyclical	1
ZBH	2026-07	Healthcare	1
ZBRA	2026-07	Technology	1
ZIM	2026-07	Industrials	1
ZM	2026-07	Technology	1
ZS	2026-07	Technology	1
ZTS	2026-07	Healthcare	1

Table 1. ESG scores/signal

metric	esg	spy
annualized_return	0.12	0.12
annualized_volatility	0.18	0.17
sharpe_ratio	0.68	0.69

Table 2. ESG portfolio performance vs SPY

Limitations

NOTE

Results are research-grade, not investment advice. The sample is limited to available data, and measured edges may not persist out-of-sample.

Appendix Tables

esg_performance

metric	esg	spy
annualized_return	0.12	0.12
annualized_volatility	0.18	0.17
sharpe_ratio	0.68	0.69

esg_scores

symbol	snapshot_mo...	sector	esgScore
1SIE.MI	2026-07	Industrials	1
A	2026-06	Healthcare	1
AAL	2026-06	Industrials	1

symbol	snapshot_mo...	sector	esgScore
AAPL	2026-06	Technology	1
ABBV	2026-06	Healthcare	1
ABN.AS	2026-07	Financial Services	1
ABNB	2026-06	Consumer Cyclical	1
ABT	2026-06	Healthcare	1
ACA.PA	2026-07	Financial Services	1
ACGL	2026-06	Financial Services	1
ACHR	2026-06	Industrials	1
ACLS	2026-06	Technology	1
ACM	2026-06	Industrials	1
ACN	2026-06	Technology	1
ACS.MC	2026-07	Industrials	1

Statistics Summary

Key	Value
study	esg_factor_backtest
category	credit
status	success
parameters	{'lookback_years': 5, 'horizon': 63, 'top_n': 50}
datasets	['fundamentals/fundamentals_esg', 'fundamentals/fundamentals_curated']
rows	12692
git_commit	db7c41b85ad2bab5b260961cfa07eff12f61aa8e
executed_at	2026-08-05T07:36:24.279864+00:00

Glossary

Term	Definition
n / count	Number of observations or events in the sample.
edge	Average excess return versus the unconditional baseline.
hit-rate	Share of events with a positive forward return.
p-value	Probability of observing the result under the null of no edge.
*** / ** / *	Significance at 99.9% / 99% / 95% confidence.
α (alpha)	Intercept or active return versus benchmark.
β (beta)	Sensitivity to a reference factor or market.

Term	Definition
σ (sigma)	Volatility / standard deviation of returns.

Document Control

Field	Value
Report schema	2
Framework version	0.1.0
Classification	internal
Generated	2026-08-05 07:36 UTC
Author	Cerberus Research
Study slug	esg_factor_backtest
Run ID	20260805_083624
Git commit	db7c41b8
Datasets	fundamentals/fundamentals_esg, fundamentals/fundamentals_curated
Parameters hash	2c2d39503ff6
Parameter · lookback_years	5
Parameter · horizon	63
Parameter · top_n	50
Data fingerprint	c699cf122334
Duration	39.5s